

KRIS KRUEGER

Engineering Lead · Software Engineer

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Software engineer and engineering lead with 4+ years building test automation platforms, internal tooling, and GUI-driven systems for mission-critical hardware programs. Experienced writing production Python, Tcl, and Bash; deploying software at scale with Ansible and Jenkins; running sprint planning and program delivery; and mentoring engineering teams.

TECHNICAL SKILLS

Languages: Python, Tcl, Bash, JavaScript, HTML/CSS, C, MATLAB

Tools & DevOps: Ansible, Jenkins, Git, BitBucket, Jira, Confluence, Agile/Scrum, CI/CD pipelines

Systems: Test asset bring-up, PC provisioning, networking, hardware-software integration, REST APIs

Frontend & GUI: Desktop GUI development (Tcl/Tk), JavaScript/HTML/CSS, large end-user application features

EXPERIENCE

Engineering Lead · Test Software Engineer - Northrop Grumman

Jul 2024 - Present

Baltimore, MD

- Deputy Product Owner and SME for a large test software platform used daily by engineers to validate flight hardware - own sprint planning, work prioritization, timeline communication, and executive reporting
- Stand up test assets end-to-end: provision and configure PCs, deploy and integrate test software, validate hardware communication, configure networking and Gigabit links, and extend the stack per asset requirements
- Design and ship features in **Python, Tcl, and Bash** for a large **GUI**-based test platform - building sub-GUIs, adding functionality, fixing bugs, and improving the end-user experience for internal engineering teams
- Deploy and scale test software across production environments using **Ansible**; gained exposure to **Jenkins** for automated deployment pipelines
- Author and execute automated acceptance test scripts verifying satellite payload processor functionality before delivery; write automation for integration tests, environmental validation, and RF range testing
- Manage the team day-to-day when the lead is out - assigning work based on individual strengths and tracking progress against program timelines
- Mentor engineers and interns on test software architecture, bench operation, scripting patterns, and best practices
- Earned multiple individual excellence awards for consistently delivering high-quality work

Systems & Integration Engineer - Northrop Grumman

Jul 2022 - Jul 2024

Baltimore, MD

- Designed and implemented an automated continuity test in **Python** for resistance measurement on newly manufactured electrical boards, replacing a manual process
- Analyzed and visualized flight equipment performance data using **Python** and **Jupyter** Notebooks; presented findings to technical leadership
- Tested, programmed, and delivered 25 RF electrical systems and embedded computers for flight use; developed and integrated new test software features for bench-level hardware integration

Software & Hardware Engineering Intern - AEVEX Aerospace

Jun 2021 - May 2022

Led R&D design and prototyping of a wireless touchscreen device for aircraft maintenance equipment; designed a LiDAR sensor mounting system with full CAD models.

Undergraduate Research Assistant - James Madison University

Jan 2019 - May 2022

Developed MATLAB and Python tools to analyze molecular spectra from space-based astrochemistry data sets.

PROJECTS & VENTURES

LEGO Reselling Business — Founder & Operator, 4 years: Python automation suite (price automation tool, sales tracker) supporting active BrickLink/eBay operation with thousands of listed items.

Low-Level C — Mandelbrot set image generator and stack data structure implementation in C; hosted on GitHub.

EDUCATION

B.S. Engineering, GPA 3.793 — James Madison University

May 2022

Minors: Mathematics, Physics, Honors | President's List 2019–2021